DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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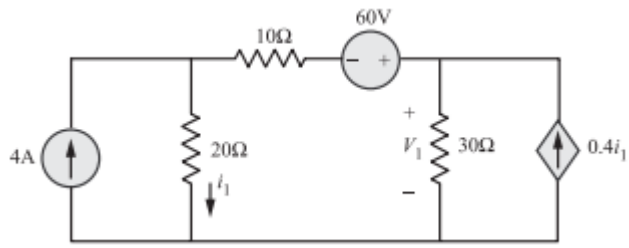
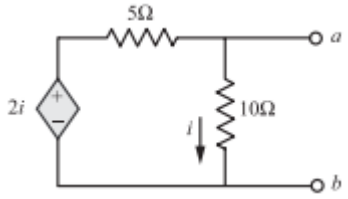
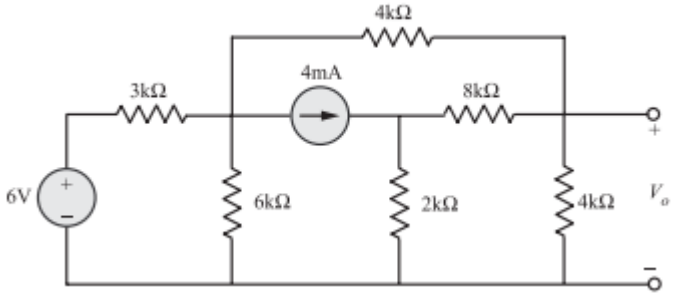
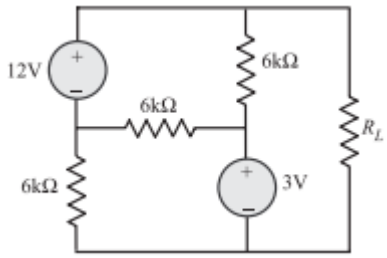
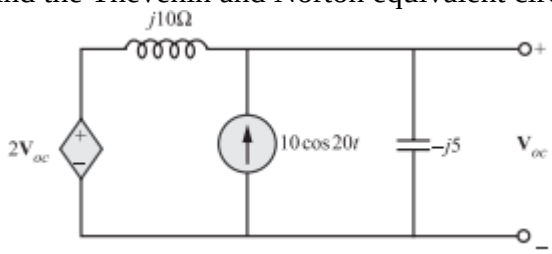
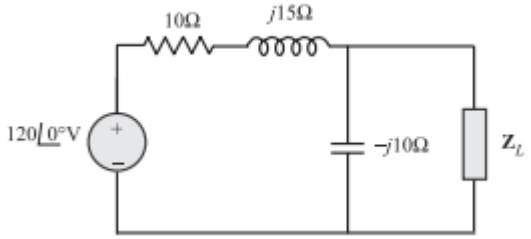
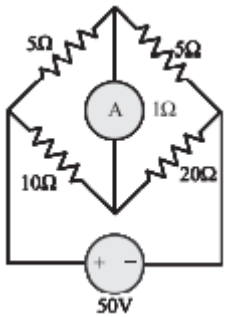
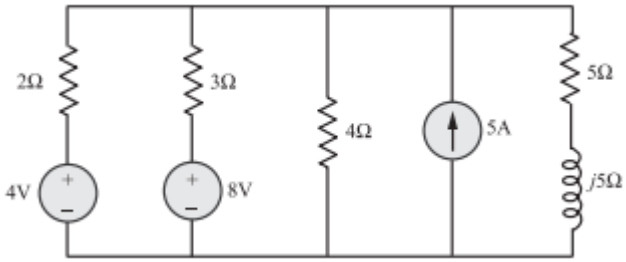
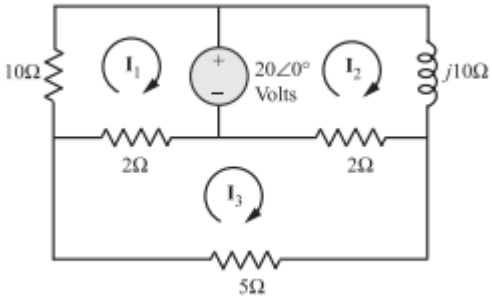
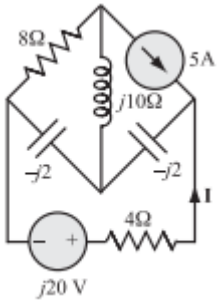
Unit 1: Tutorial 1B: Theorems

SI. No.	Problems
1	Find the voltageV1 using the superposition principle. Refer the circuit shown in Fig. 
2	Find the Thevenin equivalent circuit as seen from the terminals a-b . Refer the circuit diagram shown in Fig. 
3	Find Vo in the circuit of Fig. 
4	Find the value of R_L for maximum power transfer. Also find the maximum power transferred to R_L

	
5	Find the Thevenin and Norton equivalent circuits for the circuit shown in Fig. 
6	For the circuit, (a) what is the value of Z_L that will absorb the maximum average power? (b) what is the value of maximum power? 
7	Refer the circuit shown in Fig. Find current through the ammeter, and hence verify reciprocity theorem 
8	Refer the circuit shown in Fig. Use Millman's theorem to find the current through (5+5j) Ω impedance

	
9	Find current through 5 ohm resistor shown in Fig. and hence verify reciprocity theorem. 
10	For the network shown in Fig., apply superposition theorem and find the current I